# Securing Computer Vision Systems

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## Abstract

Algorithmic image classification has seen monumental improvement in the last decade, with neural network based techniques becoming the state of the art and widely used. However, these classifiers learn very fragile formulations of what constitutes a given class, making it possible to craft low magnitude, imperceptible alterations which greatly reduce their performance. This weakness greatly limits their practical applicability and motivates the development of robust classifiers, however usual robust classifiers require expensive adversarial training. Adversarial purification attempts solving this problem by decoupling classification from purification, and focusing on purging adversarial alterations without removing important classification features. Diffusion based adversarial purification utilizes ideas from non-equilibrium thermodynamics to model the complex distribution of clean data points. They show promising adversarial purification results but suffer from poor generalization across different perturbation types, unreliability at large iteration counts, and often falsely appear secure due to stochastic or non differentiable gradients. This report investigates improvements to diffusion based purification, suggesting and rigorously evaluating a novel approach based on frequency space guidance and fast Fourier transform truncated reconstruction. The suggested technique is based on the general observation that adversarial perturbations tend to exist in higher frequency space, while features in lower frequency space exhibit adversarial resilience. This method is tested against strong adaptive attacks assuming full white box knowledge, including model weights and exact algorithm steps. Finally, practical concerns including computation time, memory usage and scalability are analyzed, and the method is compared against existing works.

☐ **I certify that all material in this dissertation which is not my own has been identified.**

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# 1 Introduction

## 1.1 Motivation

Deep neural networks (DNNs) have dominated the computer vision space since AlexNet greatly improved the ILSVRC [1] best score in 2012, the pivot towards fully learned features has closed the performance gap between humans and computers in terms of image classification [2].

However, learning parameters only from data does not favor adversarially resilient features, just whatever features happen to minimize training loss. This generally leads to a classifier over relying on small, humanely-insignificant parts of the image, and so being vulnerable to carefully crafted perturbations.

Computer vision has found many applications across many different domains. In medicine, neural networks excel at detecting blood clots from CT scans [4] and tiny fractures from X-rays [5]. Computer vision systems can flag signs of early cancer [6] and neurological diseases such as Alzheimer's [7] long before a human doctor could. Computer vision systems are used to give robots and self driving cars the ability to understand the world around it. Yet researchers show that with just a small path of colour it is possible to fool a robotic arm into misjudging the location of a person [8], potentially injuring them. Researchers shows that the computer vision systems used in self driving cars can interpret a stop sign as a speed limit with just a few, dark patch stickers [9].

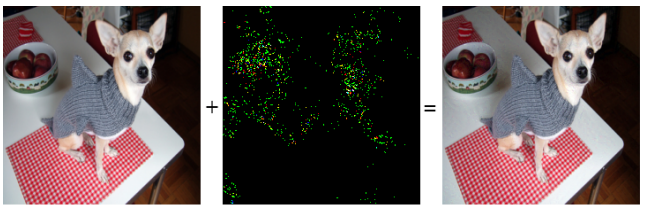



Figure 1: Adding a tiny amount of imperceptible noise to an image is enough to wildly change computer vision classification results, attack pictured is the FGSM (explained in section 4.2). Performed against a VGG16 [3] vanilla classifier on Imagenet.

Finally, the field of adversarial computer vision defense has seen rapid intellectual development in recent years. Consequently, the standard evaluation methods often fall short of accurately evaluating new methods. Many methods are designed for adversarial training, after which gradient computation is clean and accurate. Adversarial purification methods can report stochastic, chaotic, or null gradients resulting in a false sense of security.

The motivation for the proposed method and analysis is summed up below.

*Computer vision represents a promising frontier of technological integration and intellectual development, yet our state of the art methods remain vulnerable to perturbations that human beings cannot detect. Adversarial weakness, as well as the difficulty in evaluating new, novel methods severely limits the practical applicability and trust for computer vision systems.*

## 1.2 Literature and Current Methods

A conceptually simple way of combating adversarial weakness is to introduce adversarial examples to the training set (Adversarial Training - AT), and this remains the state of the art method on paper.

Bartoldson et al used a model hundreds of times larger than required for standard classification, as well as 300 million synthetic data points generated with an elucidating diffusion model [10] to increase State Of The Art (SOTA) CIFAR-10 adversarial accuracy from 66% to 74% [11] at great computational expense. Wu et al suggest directly perturbing parameters during training, and training for a low magnified change in further layers [12], a method which was quickly adopted my many SOTA frameworks.

However, the adversarial data points have to either be pre-generated, or generated alongside training. Pre-generating adversarial data points results in almost no generalization to different attack types [13]. Generating adversarial data points alongside training either massively increases training time, or results in less than state of the art adversarial accuracy. Recent developments allow for the

reuse of gradient computation graphs from parameter optimization to compute loss gradients for adversarial attacks [14]. However, one loss gradient evaluation per network parameter update simply is not enough to generate strong enough perturbations for SOTA adversarial accuracy; and evaluating gradients many times per batch proportionally increases training time.

Adversarial training also inevitably reduces clean accuracy [15], and modern adversarial accuracy results tend to require significantly more data and network parameters than regular classification [16] [17]. Adversarially trained networks also tend to overfit to perturbation type [18] [19], and allowed perturbation set geometry [20]. Novel methods also tend to beat previous generation resilient models [21] [22].

Adversarial purification instead pre-processes inputs before they are classified, decoupling the difficult problem of adversarial security. Any classifier can then be used, avoiding complex and expensive re-training. Adversarial purification utilizing diffusion, first proposed by Nie et al [23], attains state of the art performance [24] [25] [26], likely due to the inherent ability of diffusion to model complex data distributions.

Diffusion models work by learning the reverse of a degenerative process [27], which is performed on a complex distribution until it resembles a tractable distribution. By starting from this tractable distribution, the degenerative process can be reversed by subtracting data not belonging to the complex, learned distribution [28]. This process directly mirrors adversarial purification; removing out of distribution (adversarial) data while retaining clean information.

However, the strong generative power of diffusion can lead to loss of semantic content during extended runs, motivating guidance. Wang et al suggest first diffusing the adversarial image forward, to 'drown out' adversarial noise, then using it as guidance to keep the diffusion process on track [29]. Lei et al instead argue for conditioning on a latent representation of an unlearn-able, edge detection filter [30], while Li et al suggest conditioning on an intermediate denoising step [31]. Conditioning on the adversarial image, even after a forward [29] or backward [31] pass risks feeding the perturbation back into the generative process.

A strong adaptive attack can take advantage of this, as shown by Kassis et al reducing adversarial accuracy on Cifar10 to 19.53% for $\ell_\infty, \epsilon = 8/255$. This was achieved with expectation over transformation (EOT) sampling [32], static random seeding and careful error accumulation control.

A significant weakness of diffusion based purification is its indifference towards features of different frequency bands; adversarial perturbations tend to live in high frequency, 'noisy' space [33] [34], while low frequency data retains global structure after an attack [35]. Pei et al suggest performing a discrete Fourier transform (DFT) on an adversarial image, decomposing it into phase and amplitude spectra, then performing reverse diffusion only on high frequency components, keeping adversarially resilient low frequency bands from the perturbed image. While this increases clean and adversarial accuracy, it is vulnerable to perturbations specifically crafted to fit the low frequency band, as exploited by Kassis et al [36].

An inherent weakness of diffusion based defenses is the computational cost of performing reverse diffusion, which is notoriously compute hungry [37] due to large neural network evaluations per denoising step. Significant work has been done to improve upon this bottleneck, Li et all suggest using optimized samplers such as DDIM [38], as well as fine tuning the diffusion model on adversarial noise [24] to achieve effective purification in fewer steps. Lei et al suggest training a smaller surrogate model to map latent representations of adversarial data points to clean ones, achieving close to SOTA performance on CIFAR-10 in a single digit number of steps [30].

# 2 Project Specification

## 2.1 Aims and Objectives

The project is built around 5 high-level objectives, each with specific, measurable success conditions. Aim 1 addresses the theoretical design of the proposed algorithm, focusing on justifying each design

decision with literature or custom experiments. Aim 2 practically tests against standard white-box attacks and aim 3 tests against advanced adaptive attacks. Aim 4 focuses on comparing against similar methods and properly establishing limits of the proposed method.

The general goals of this project are summarised below:

*To develop an efficient adversarial purification scheme that works independently of perturbation type, is consistent at large iterations and adapts well to unseen circumstances. The scheme is to require no adversarial training, fine tuning or large scale parameter optimization outside of the original classifier and diffusion model. The aim is to produce an algorithm that forces perturbations into scales and shapes that are detectable by normal people, no longer making it possible to consistently adversarially perturb images in a way that is humanely indistinguishable.*

This is achieved by accomplishing the objectives in table 1

Table 1: Research aims and objectives

| | |
|---|---|
| **Aim 1: Design a diffusion-based adversarial purification framework** | |
| Obj: 1.1 | Develop an adversarial purification algorithm based on diffusion, frequency space processing and classifier free guidance which is effective under full white-box threat assumptions; the attacker has complete knowledge of all model parameters and exact algorithmic steps. Each design choice is to be justified with supporting literature or custom experiment results. |
| Obj: 1.2 | Implement the proposed algorithm in PyTorch, using pre-trained CIFAR-10 and MNIST classifiers. This is to result in an automatic pipeline which classifies potentially adversarial data points. |
| Obj: 1.3 | Investigate different surrogates for each step in the proposed pipeline, rigorously evaluating each one against multiple attack types. |
| **Aim 2: Achieve strong robustness against standard white-box attacks** | |
| Obj: 2.1 | Achieve $\geq$80% clean and robust accuracy against FGSM [39] and PGD attacks at $\ell_\infty$, $\epsilon$=8/255 and $\ell_2$, $\epsilon$=0.5 on both CIFAR10 and MNIST. |
| Obj: 2.2 | Achieve $\geq$60% clean and robust accuracy against The Carlini-Wagner (C&W) [21] and Square [40] attacks. If an attack does not allow a forced $\epsilon$ norm bound while retaining strong performance, success is defined as the measured difference exceeding an $\ell_2$ norm bound of 0.75. Perturbations over this magnitude are no longer imperceptible. |
| **Aim 3: Evaluate robustness under advanced adaptive attacks** | |
| Obj: 3.1 | Achieve $\geq$60% robust accuracy against PGD + Backward Pass Differentiable Approximation (BPDA) + EOT on CIFAR-10 at $\ell_\infty$, $\epsilon$=4/255 and $\ell_2$, $\epsilon$=0.2. |
| Obj: 3.2 | Achieve $\geq$60% robust accuracy against C&W + BPDA + EOT under the same norm bounds as above. |
| Obj: 3.3 | Verify robust accuracy against perturbations concentrated in high frequency bands, ensuring adaptive attacks do not achieve better than $\geq$60% robust accuracy. |
| **Aim 4: Directly compare against similar algorithms** | |
| Obj: 4.1 | Compare the proposed method with at least 2 similar published works, specifically outlining where the proposed method performs better or worse. |
| Obj: 4.2 | Achieve better robust accuracy results than DiffPure [23] against standard and strong attacks. |

### 2.1.1 Deliverables

- A detailed description of the proposed adversarial purification framework, with each detail and choice extensively justified with either proven results or supporting literature.
- Comprehensive numerical results covering all attacks, norm bounds and datasets specified in Table 1 under aims 2 and 3.
- Static comparison against published diffusion-based adversarial defence algorithms, quantitative comparison results against DiffPure [23], FreqPure [35] and GDMP [29] methods.

- Critical analysis of the proposed method strengths and weak points, including per-attack reasons for why certain attack modes perform better or worse. All backed by theoretical and practical justifications
- A parallel, efficient Pytorch implementation of the described framework.

### 2.2 Stretch Goals

- Achieve better results that FreqPure [35] and GDMP [29] against a set norm bound.
- Test against attacks which generate low frequency 'smooth' perturbations that evade traditional high frequency filtering.
- Evaluation against other datasets, CIAFR-100 or Fashion-MNIST to test generalization to other data distributions.
- Test diffusion model zero shot purification performance. The bottleneck of the practicality of the proposed method is training a diffusion model that accurately models the clean data distribution, however diffusion models have demonstrated remarkable zero shot performance in other areas [41] [42], perhaps purification may be partially transferable skill.

## 3 Design and Implementation

This section details the proposed purification algorithm. The system utilizes three core mechanics: (a) frequency space denoising via 2D discrete Fourier transform deconstruction and truncated reconstruction, (b) frequency aware, classifier-free guidance performed without the need for a conditionally trained diffusion model, (c) logit level majority voting across cleaned scheduler predictions.

The required theoretical basis of each element is briefly explained, followed by an in depth justification of each choice compared to similar methods, backed up by published literature or experimental results.

A high level overview of the algorithm is given below:

1. **Noise and $x_0$ Prediction.** Initially, the potentially adversarial data point $x$ is passed through a diffusion neural network, which predicts which pixels do not follow the distribution it was trained on. Utilizing a DDIM [38] sampler, a prediction of the image after many purification steps, $x_0$ is returned at no extra compute cost.
2. **Biasing for Long Run Consistency** To allow many diffusion iterations without accuracy loss, $x$ is quantized to get $\hat{x}$. $x_{guide}$ is computed by taking low frequency, structural data from $\hat{x}$ and high frequency, noisy data from $x_0$. The noise prediction is biased towards $x_{guide}$ using an analytic simplification of the guidance scheme devised by Wang et al [43] to avoid the requirement of a conditionally trained diffusion model.
3. **Denoising** $x$ is de-noised a single step using the biased noise prediction to get $x_{biased}$, which is passed onto the next diffusion iteration.
4. **Majority Voting** Finally, the collection of $x_{guide}$ values computed during intermediate denoising steps is passed through the classifier back-end, and a majority vote is performed on the resultant logits. Later $x_{guide}$ values are weighed more heavily as the signal-noise ratio of the scheduler converges.

The general intuition behind this algorithm goes as follows. Initially the out of distribution data of the image is predicted with a diffusion network, using a DDIM sampler a prediction of the mean of all possible final clean images is computed $x_0$ at no extra cost. The diffusion process is not adversarially

perfect, and carefully crafted perturbations can evade the noise prediction, specifically perturbations which follow natural patterns in the image, as supported by the work of Chen et al [44] as well as Kassis et al [36]. Strong adaptive attacks can generate such perturbations and fool the diffusion model.

In order to avoid this problem, the final image after many timesteps is never classified, instead at every step as much data is taken from the $x_0$ predictions as possible without compromising robustness.

Inspired by the work of Carlini et at [45] who perform Gaussian smoothing on $x_0$ estimates to achieve certified robustness bounds, the proposed method performs a series of operations which attempt to purge both high and low frequency alterations. First $x$ is quantized to get $\hat{x}$, many low frequency alteration resistant functions were tested and quantization performed the best on average across norm bounds and attack types, this choice is further justified in section 3.3.1. The useful $\hat{x}$ low frequency data is combined with the high frequency data from $x_0$ utilizing a 2D Discrete Fourier Transform based process heavily based on FreqPure by Pei et al [35].

The computed $x_0$ represents the mean of many possible final states and so is naturally resistant to high frequency alterations.

The combination of low and high frequency data from mean states, and heavily quantized outputs however does not produce good accuracy alone, necessitating the majority voting scheme. Early $x_0$ predictions are of lower quality as not many denoising steps have been performed yet, so the $x_{guide}$ values are less predictive and so early votes are lower weighed. The standard diffusion formulas provide a natural way to measure $x_0$ sample quality: Signal Noise Ratio (SNR). During forward diffusion, the clean data point is degraded by performing a weighted sum between clean data and Gaussian noise. The ratio of contribution of noise and clean data at a specific timestep is called the signal-noise ratio, and naturally represents current sample quality.

## 3.1 Frequency Space Processing

The general observation that adversarial perturbations tend to have high frequency, noisy components has been echoed throughout adversarial literature [33] [35], with recent work arguing that the lower the resolution the dataset, the higher frequency perturbations tend to be [46].

It is possible to explicitly optimize for low frequency perturbations [47] [48] however doing so is more difficult within common norm bounds. An $\ell_\infty$ bound strongly encourages uniform perturbations as each pixel has a set allowed range, $\ell_2$ bounds penalize by the square of euclidean distance, so large smooth alterations are heavily penalized. The less used $\ell_1$ norm bound allows for such alterations, but results in clear visual degradation. Convolutional Neural Network (CNN) based classifiers also heavily rely on high frequency data, making them generally less sensitive to low frequency alterations, as justified in section **??**. Standard gradient based attack methods are also usually biased towards generating high frequency perturbations, specifically PGD and FGSM [49], and optimizing for low frequencies tends to be more computationally expensive and algorithmically difficult than operating directly on pixels [50].

Though these limitations are significant, the proposed method must still be secure against low frequency perturbations, which is addressed in section 4.3.

Generally, it is useful in the context of adversarial purification to be able to separate the high and low frequency data contained in an image. As justified, adversarial perturbations are biased towards high frequency components, and different classifier architectures respond differently to low and high frequency data. Practically, frequency space decomposition makes it possible to perform reverse diffusion on only vulnerable data, preserving important global semantic information [35]. Guidance is also far less vulnerable when made frequency-aware.

### 3.1.1 Frequency Space Deconstruction and Reconstruction Methods

The described frequency band blending can be achieved in several ways, such as rowPCA [33], introduced to defend against the C&W [21] attack. RowPCA first reshapes the image by folding a spacial dimension with the colour channel dimension into a 2D matrix, then decomposes it as the sum of two

components, one sparse, one low-rank. Natural image data is better represented by the low rank component as it contains smooth structure with large spacial correlation, while adversarial data consists of small scale, sparse, per-pixel deviation, being naturally represented by the sparse matrix. Combining low-rank and sparse matrices from two image achieves the desired frequency band blending.

Another way of achieving this goal is to use a 2D discrete Fourier transform (DFT), explicitly operating in frequency space. The DFT decomposes the data into a sum of sine functions of different magnitudes and phases, represented by a complex valued matrix and each one mapping to some global element of the image. Low spacial frequencies, those near the centre of the computed spectrum (the complex values matrix) correspond to smooth gradients and broad shapes while those near the edges encode finer detail and edges. By using a euclidean distance filter, it is possible to combine frequencies from two images, also achieving the desired band blending.

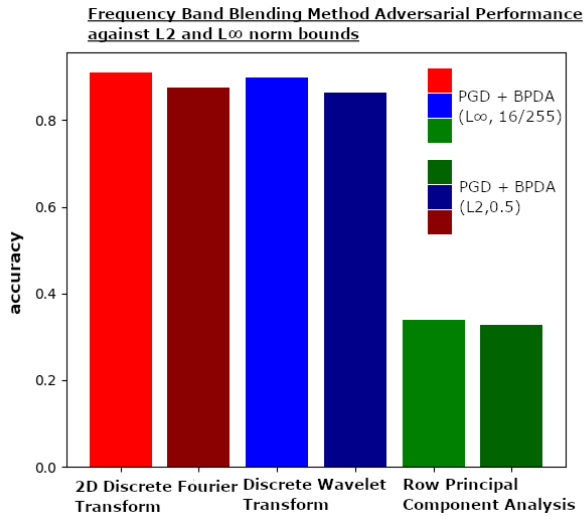



Figure 2: 40 step Projected Gradient Descent attack results against n=512 CIFAR10 data points (repvgg-a2 classifier). 30 diffusion steps used for all attacks.

A seemingly superior way of achieving this goal is using a Discrete Wavelet Transform (DWT), a local version of the DFT. The 2D DFT assumes that each component is present globally over the image, so each pixel contributes to each computed phase and amplitude. A DWT on the other hand uses a basis function that is local in both space and frequency, so instead of sine waves that cover the entire image, DWT uses small local 'wavelets'. A DWT also completes in $O(n)$ compared to $O(n \log n)$ for a DFT (with respect to number of pixels $n$).

In testing however, DWT based purification seems to attain strictly worse results across norm bounds, as demonstrated in section 2 though the difference is small, 1%-3%. This may be due to its local properties, an adversarial perturbation is often spread over large parts of an image as small, per-pixel changes. Encoding this with a global basis function aggregates many individual pixel changes into a small number of frequency bands, making filtering more efficient. Encoding this data with a local basis function distributes its potential across many resultant coefficients, less cleanly dividing adversarial from clean data after filtering. However both DWT and DFT are used in recent adversarial purification literature, so the poor performance in this context is probably due to the simplicity of the filtering scheme.

Ultimately, 2D DFT based decomposition and truncated reconstruction was chosen as it was most adversarially performant, took less time to compute and parallelised better on a GPU.

Formally, $FFT(x_{\text{low}}, x_{\text{high}})$ denotes the image reconstructed by taking low frequency components of image $x_{\text{low}}$ and high frequency components of image $x_{\text{high}}$.

$$\phi_{low}, \phi_{high} = \mathcal{F}(x_{low}), \mathcal{F}(x_{high})$$

$$\phi_{comb} = \phi_{low} \odot M + \phi_{high} \odot (1 - M), \quad M_{u,v} = \begin{cases} 1 & \sqrt{u^2 + v^2} \leq r_{\text{cutoff}} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{comb} = FFT(x_{\text{low}}, x_{\text{high}}) = Re(\mathcal{F}^{-1}(X_{comb}))$$

*$\mathcal{F}(x)$ represents the 2D DFT decomposition of $x$ into phase and amplitude spectra $\phi$. Element-wise multiplication is represented with $\odot$ and $M$ represents the mask used to filter high and low frequency*

*data. $Re(z)$ represents the real part of complex number $z$.*

The frequency band blending described is used frequently in the proposed method, and is generally denoted with $FFT(A, B)$. All frequency band blending is 2D DFT based as justified above.

## 3.2 Diffusion for Purification

### 3.2.1 Diffusion Background and Motivations

In 2017, Carlini et al introduced a simple but effective adversarial attack scheme, the C&W attack. Explained in section 4.2, this attack broke many adversarial purification schemes. Utilizing the ADAM [51] optimizer, a culmination of recent advancements in optimizer technology, as well as a loss function that targets the second most likely logit, it rendered many simpler purification methods obsolete. Most notably randomized smoothing, blurring and normalization based defences, as well as PCA based defences [52] and methods based around detecting if a data point was adversarial were crippled. This is supported by my own experiments, when using randomised smoothing to guide the diffusion process, the C&W (+ BPDA [22]) attack managed to reduce adversarial accuracy down to 54.67% on CIFAR10.

The C&W attack became the standard for testing adversarial purification schemes, and so simple methods just based on perturbation detection or obfuscated gradients were shown to provide little adversarial robustness. This motivated a new generation of adversarial defence which attempt to model the clean data distribution and project data points back into it. Variational AutoEncoders (VAEs) and Generative Adversarial Networks (GANs) were first used to achieve this [53] [54] to varying success. VAE defences require expensive adversarial training and fall to 44%-50% accuracy under strong adaptive attacks [53], with their latent compression often being simple enough that a strong attack can craft a perturbation that survives. GAN based defences are very difficult to train, with it being difficult to avoid mode collapse, they also scale poorly to complex data distributions [55] and tend to suffer manifold gaps [56] which attack algorithms excell at taking advantage of. Diffusion models generally converge better during training than GANs, have better reconstruction quality than both GANs and VAEs [25] and have been shown to scale better to more complex problems.

Diffusion models generally define a forward degenerative process that corrupts data from a complex distribution until it resembles a simple distribution. Practically, the relevant complex distribution considered is one of unperturbed images for classification and the degenerative process is adding Gaussian noise over $T$ timesteps.

Formally, Gaussian noise of mean $\mu$ and variance $\sigma$ is repeatedly added to an image $x_{noisier} = N(x_{cleaner}; \mu, \sigma)$. The process is represented as a Markov chain with learned Gaussian transitions [28]. $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$, with $\beta_1 ... \beta_T$ being a variance schedule controlling how harshly data point $x$ is corrupted at time step $t$. A neural network is then trained to predict the noise $\epsilon$ added at each time step by optimizing for the evidence lower bound (ELBO) objective: $\mathcal{L} = \mathbb{E}_{x_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$ Where $\epsilon_\theta(x_t, t)$ denotes the neural network noise prediction under parameters $\theta$, and $\mathbb{E}_{x_0,\epsilon,t}$ denotes expectation over all clean images $x_0$, Gaussian noise permutations $\epsilon$ and schedule timesteps $t$. The model learns to predict data that is not in its unperturbed distribution, directly matching the goals of adversarial purification.

Diffusion models now dominate most adversarial image purification schemes [56] [30] for reasons discussed above, and so diffusion was chosen as the back-end of choice for the proposed method.

### 3.2.2 Practical Diffusion Details

Though many methods exist to sample a reverse diffusion process, generating data of the learned distribution, DDIM [38] was chosen as it computes $x_0$, an estimate of the image after many denoising steps, at no extra compute. Practically, $x_0$ is computed with Tweedie's formula [57] and just performing majority voting on each $x_0$ estimate after each denoising pass is enough to get far better results

than DiffPure [23] against weak to medium level attacks. This alone is enough to greatly improve upon DiffPure against simple attacks (see Figure 4.2)

Spectral analysis performed by Pei et al [35] demonstrates that this $x_0$ estimate tends to be overly smooth, more representing the average of possible clean images, this is supported by Song et al [38], the authors of the DDIM sampler used. This is useful as $x_0$ naturally tends to cancel out the high frequency perturbations introduced by attacks, Pei et al use this cleaned high frequency data to replace adversarial data at each step of the diffusion process.

Instead, I chose to use this high frequency data to construct $x_{guide} = FFT(q(x_{adv}), x_0)$ to guide the diffusion process and pass off to the majority voting system. Here, $q$ represents quantization, a choice explained in section 3.3.1. I chose to pass $x$ after a single diffusion step to the next iteration without any frequency space blending as this resulted in a generally more stable diffusion process. It allowed the use of stronger, more adversarially resistant guidance scheme that was far more resistant to low frequency perturbations, this is further justified in section 3.2.3. Not performing frequency band blending directly on intermediate denoising steps also helped prevent potential perturbations from persisting throughout the purification chain. Generally, the authors of FreqPure [35] attempt to maximize the amount of useful information they can get out of the adversarial image by directly using its low frequency components at each iteration. I instead allow the imperfect diffusion process to continue uninterrupted, but at each step heavily guide the diffusion towards poor but adversarially resistant transformations, keeping resistant data for majority voting later. The effectiveness of both approaches is directly compared in section 4.4. I found that while performing intermediate frequency band blending does improve both robust and clean accuracy, it is incredibly sensitive to noise injections and guidance. While a certain set of hyperparameters would do better against one attack, the performance would not transfer to another attack type or norm bound without hyperparameter fiddling, defeating the point of having one purification scheme that works independently of attack type.

### 3.2.3 Guided Diffusion

Diffusion is an inherently generative process, and so extended runs will overwrite important data. Extended runs are however crucial for defending against strong adaptive attacks, for example a Ultimate-PGD + BPDA + EOT attack requires at least 100 diffusion operations to get good defensive results, as shown in section 4.2. Guiding diffusion can be done in several ways, it was initially done by using external classifier gradients to bias each noise prediction [58], however this process would not work in this context, as said classifier could be fooled by a strong adaptive attack. State of the art guidance involves injecting information about the image when training the diffusion model, sometimes providing null information so the model learns specific as well as general conditions. However conditional training is computationally expensive, and this project aims to produce a method which requires as little extra computation as possible.

Instead, I opted for a heuristic guidance scheme similar to Wang et al [29], which works by reformulating the unconditional distribution $p(x_{t-1}|x_t)$ to also depend on a $x_{target}$ term. Note that these distributions also depend on the set of current parameters, $\theta$, as this is required to predict noise, this is left out of the formulas for brevity.

Applying Bayes rule and rearranging gives the following. $p(x_{t-1}|x_t, x_{target}) \propto p(x_{t-1}|x_t) \cdot p(x_{target}|x_t)$ Taking logs gives:

$$\log p(x_{t-1}|x_t) + \log p(x_{target}|x_t) + K_1$$

Where $K_1$ is the constant from taking the log of a proportionality relationship. Crucially, $p(x_{target}|x_t) = \int p(x_{target}|x_{clean}) p(x_{clean}|x_t) dx_{clean}$ is computationally intractable as it would require marginalizing over the entire denoising trajectory. The log of the integral is therefore approximated with a maximum entropy distribution under a distance constraint [29] to get the following approximation.

$$p(x_{target}|x_t) \approx \exp(-sD(x_t, x_{target}))$$
$$\mu_{\text{new}} = \mu(x_t, t) - s\sigma\nabla_{x_t}\mathcal{D}(x_t, x_{\text{target}})$$

Where $s$ is a guidance strength hyperparameter, $D$ is a distance metric and $\sigma$ is the current variance output. The given term is then added to the mean of every noise prediction $\mu$, biasing both the previous denoising step $x_{t-1}$ as well as the $x_0$ prediction to be used in the next time step.

Practically, without guidance clean accuracy falls off very fast as diffusion step size increases, as shown in Figure 3

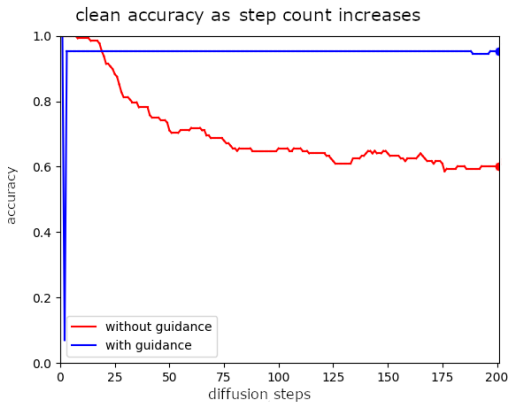



Figure 3: Clean accuracy as number of diffusion iterations increases, performed on CIFAR10 using the Resnet-56 [59] classifier. Note that the initial decrease in accuracy is from the initial noise injection, the guidance scheme allows accuracy to quickly recover.

Wang et al use autograd [60] to compute $\nabla_{x_t}\mathcal{D}(x_t, x_{\text{target}})$, allowing the use of any differentiable distance metric, they find that the Structural Similarity Index Measure (SSIM) got the best results on average. However, in order to evaluate against strong adaptive attacks, a gradient of the entire proposed method must be calculated. Computing the gradient of a gradient results in autograd having to store large second derivative Hessian tensors in memory, which scale quadratically with respect to input size. This practically limited me to only processing 1-4 3x32x32 images at a time at 12gb of video memory.

Instead, it is possible to use euclidean distance as the distance metric and analytically simplify this, resulting in slightly worse performance but linear memory usage with respect to input size. Using this analytic simplification bypasses the double gradient calculation from autograd. Utilizing forward universal guidance as formalized by Bansal et al [61], computing the required gradient from the $x_0$ estimate instead of from intermediate denoising steps $x_t$ tends to results in smoother and more stable guidance. This also has the benefit of natural resistance to higher frequencies, the benefit of which is discussed in section 3.1. The analytic simplification is given below.

$$D(x_0, x_{\text{target}}) = \frac{1}{2}\|x_0 - x_{\text{target}}\|^2 \implies \nabla_{x_0}\mathcal{D} = -(x_{\text{target}} - x_0)$$
$$\nabla_{x_t}\mathcal{D} = \frac{\nabla_{x_0}\mathcal{D}}{\sqrt{\bar{\alpha}_t}} = -\frac{(x_{\text{target}} - x_0)}{\sqrt{\bar{\alpha}_t}}$$
$$\mu_{\text{new}} = \mu_\theta(x_t, t) + s_t \frac{\nabla_{x_t}\mathcal{D}}{\|\nabla_{x_t}\mathcal{D}\|}$$

Wang et al do not perform $\ell_2$ normalization here, they instead tune $s_t$ to match the scale of the gradient, while this approach gets the best theoretical performance, it would require re-tuning $s_t$ for different attacks. This would defeat the point of having a transferable algorithm that adapts to different attack situations, opting for normalization here keeps the gradient magnitude within reasonable bounds without tweaking.

## 3.3 Misc Implementation Details

### 3.3.1 Choice of Low Frequency Perturbation Resistant Function

As explained in section 3.1, adversarial perturbations tend to live in high frequency space, however the designed method must be resistant against any perturbation types. For example, Guo et al use an inverse discrete cosine transform to map pixels directly to low frequency coefficients, optimizing in frequency space and fooling the Google Cloud vision api [47]. Adaptive attacks such as the C&W [21] attack will naturally optimize for a lower frequency perturbation if that is what fools the system.

In order to ensure security against these adaptive attacks, the proposed method must be resilient to adaptive attacks. This was achieved by a performing a low frequency perturbation resistant transformation on the adversarial image before using its low frequency data to compute $x_{guide}$. Several functions were tested, Local Contrast Normalization (LCN), a differentiable approximation of Contrast Limited Adaptive Histogram Equalization (CLAHE), randomized Gaussian smoothing. LCN and CLAHE however, could not be used for majority voting as they shifted the pixel values of the images too far away from the original distribution and no longer provided a clean accuracy. They could still be used for guidance however and saw varying success at generating guidance targets, as shown in figure 4 where each potential function was tested against the C&W attack [21]. The $\ell_2$ norm bound C&W attack was chosen as it naturally tends to produce low frequency perturbations, and is good at adapting to different circumstance. The attack was adapted to produce $\ell_1$ bound perturbations similar to Chen et al [62], by adding a sparsity penalty term $||\delta||_1$, in this case the sum of absolute terms worked well. The $\ell_2$ loss term is kept to keep the optimization convex, and generally more stable.

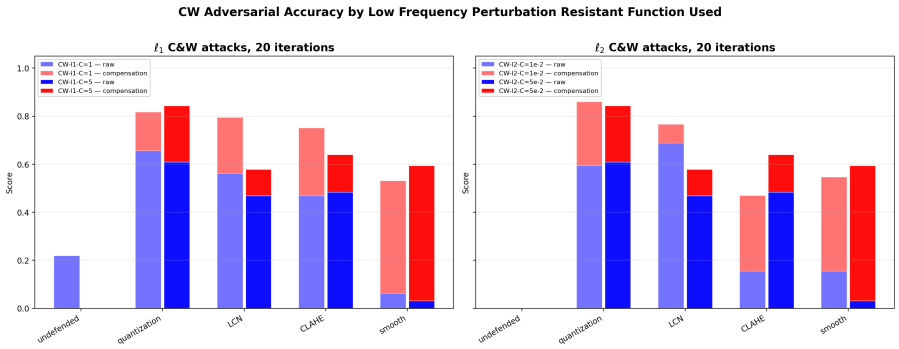



Figure 4: C&W [21] attack performance using different low frequency perturbation resistant functions. 100 diffusion steps used for all attacks. Blur parts of the bars represent the data points that were not able to be perturbed after 20 iterations, the red are the data points that were perturbed, but their $\ell_2$ norm bound exceeded 0.75. Tested against the same subset of 512 CIFAR10 images using a ResNet-56 [59] classifier post purification.

Local contrast normalization had better strict performance against $\ell_1$ norm bound perturbations: disregarding disqualified data points. In all other areas, quantization comes out on top. Interestingly using randomized smoothing resulted in almost all data points being perturbed, but nearly all of them were forced into clearly visible bounds, the results were worse than the other choices in all other regards however.

Quantization was chosen as it performed the best on average, and the output was close in terms of absolute pixel value, so this term could be used for more than just guidance.

Quantization is non-differentiable at rounding boundaries, and so BPDA with an identity backwards surrogate was used to reduce overly optimistic results. While quantizing inputs would usually greatly reduce clean accuracy, the majority voting makes this reduction negligible as long as quantization level is tuned to be the right level for the number of voting steps used. This balancing, as well as further hyperparameter choices are discussed in section 4.1.

| | full algorithm | without guidance | without frequency space blending |
|---|---|---|---|
| 30 step PGD | 0.812 | 0.75 | 0.375 |
| 5 step PGD | 0.709 | 0.71 | 0.175 |

Figure 5: Ablation Study, each isolatable element is toggled and performance is evaluated against an $\ell_\infty, 4/255$ norm bound.

Performing an ablation study (Figure 5 against $\ell\infty, 4/255$ norm bound 50 step PGD verifies that each proposed element meaningfully contributes towards adversarial accuracy. Performing the same test against clean accuracy, however, exposes that the further improvements to DiffPure + $x_0$ voting actually decrease clean accuracy .

# 4 Experiments

Rigorously evaluating adversarial purification schemes can be difficult, as it is not always clear that a good adversarial accuracy score is due to actual security and not accidental obfuscation. Carlini et al [22] showed that many seemingly secure adversarial defence schemes only appeared secure due to non-differentiable or stochastic elements, breaking through 8 out of 9 schemes testing by making linear assumptions and compensating for randomness. Generally, many adversarial attack algorithms have been designed for use against adversarial training, where arbitrary gradient computation is accurate and usually deterministic.

Formally, the proposed method has a non-differentiable quantization step and utilizes reverse diffusion, which is notorious for having exploding and non reliable gradients. If a standard gradient based attack method were tested without any alterations, it would record a very optimistic score as the gradient data required (usually $\frac{\partial image}{\partial loss}$) would be of low quality. However, the proposed method is to be secure in a white box setting, that includes an attacker isolating every non-differentiable element and artificially estimating its gradient better than whatever default is used. This described backward pass differentiable approximation (BPDA) [22], a necessary addition to gradient based attack algorithms to ensure non-biased evaluation. Formally, BPDA allows the forward pass of a non-differentiable element to carry through uninterrupted, but for the backward pass the gradient is estimated with some surrogate.

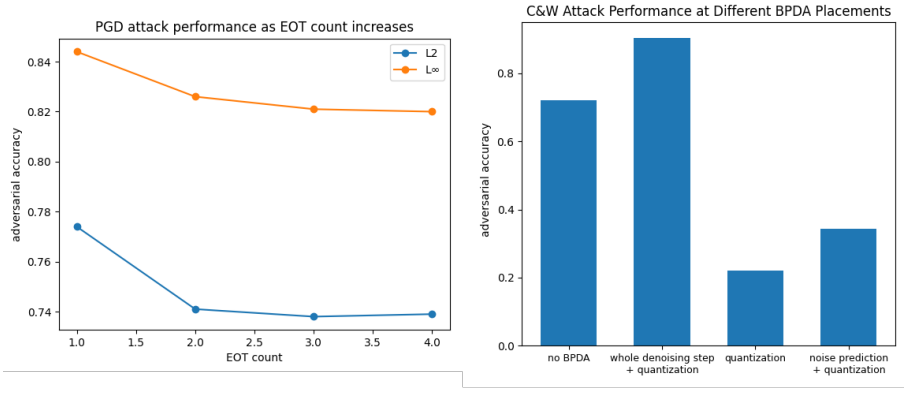



Figure 6: LEFT: PGD + BPDA + EOT attack results against 10 diffusion steps, RIGHT: 100 step C&W [21] attack against only 10 denoising steps. (Higher bar means attack performed worse given each BPDA placement), both tested against n=512 CIFAR10 data points

Generally there is no single correct method for performing BPDA, and in order to claim white box adversarial resilience, it has to be shown that there is no possible surrogate function placement that generates valid perturbations. For this reason, many different surrogate placements were tested, and the most effective used for further testing. To test different placements, a 100 step C&W [21] attack was performed against the algorithm utilizing only 10 denoising steps, and no noise injection or stochasticity. Note that these parameters are not strong enough to defend against this

attack, this was done on purpose to exaggerate the effect of each placement and compare them.

Identity BPDA was placed over the quantization for every test, as this seemed to strictly increase attack performance. Other placements were not so obvious, in diffusion purification literature, BPDA is usually placed over the entire denoising process. However, in this case doing so actually made the attack perform worse (second bar Figure 6) than no BPDA. This is likely because usually BPDA is used to avoid gradient signal dying over a long diffusion chain before the classifier, but in this case each iteration influences each $x_{guide}$ computation, which gets passed onto the majority vote so the gradient signal stays strong. This explains why no BPDA performs better, as forcing an identity gradient over a differentiable process results in information loss and therefore a weaker attack. Other literature suggest placing BPDA over just the noise prediction element, doing so increased attack effectiveness, but not as much as only using BPDA only for quantization. This for the same reason as before, there must be tractable gradients in the noise prediction which are being purged by BPDA.

Several backwards surrogate functions were briefly tested, including a DiffAttack [63] style estimator based on the diffusion model score function, and scheduler variance scaling. Ultimately, these increased compute without a measurable increase in adversarial performance over a simple identity mapping. This supports previous reasoning, the gradients through the denoising process must contain useful information that is purged by BPDA surrogates. Other works have found success with these methods though [63], the failure of more complex surrogates is probably due to the simplicity of the proposed method compared to state of the art diffusion purification methods.

The proposed method also is stochastic in several places, random noise is added during the forward pass to attempt to drown out perturbations and the reverse denoising process is random as it is estimating the solution to a stochastic differential equation. As the output of the purification process can be random, so can its gradient, and so attack methods that depend on a reliable gradient direction can fall short. This motivates Expectation Over Transformation (EOT) compensation, which involves performing a gradient computation step multiple times and averaging, to find a direction independent of noise. However, this is expensive, as it involved backpropagating through the entire diffusion chain multiple times per gradient computation, proportionally increasing running time. To compensate for this, random sources were fixed wherever possible. A noise tensor is generated at the start of the diffusion loop and reused wherever possible and the stochasticity parameter for the DDIM sampler was set as low as possible without compromising on adversarial performance.

Next, testing was performed against multiple norm bounds to find the point of diminishing returns of increasing EOT iterations. Tests were conducted with standard PGD against $\ell_2$ and $\ell_\infty$ norm bounds, increasing EOT from 1-4 as shown on Figure 6. Attack performance stopped increasing by 4 iterations, with marginal gains over 3 iterations, so and EOT of 4 was used whenever needed. Note that the results stopped increasing after only 4 iterations as great care went into reducing sources or randomness wherever possible as explained above.

### 4.1 Hyperparameter Choice

A significant goal of this project is to design an algorithm that works against a multitude of different attacks and allowed bound types, this makes hyperparameter choice difficult. It is possible to tune hyperparameters in a way that make the proposed method work well against one problem set only for it to fail against another. Generally, authors of literature will tune hyperparameters to look good for a certain problem set to show the upper potential of their methods, this however goes against the goal of a generalizable method.

Most difficulty in tweaking hyperparameters comes from balancing the amount of noise added before diffusion, the quantization level and the number of diffusion timesteps. Generally, higher levels of noise added and lower quantization levels greatly improve adversarial accuracy at the cost of clean accuracy. This is a major drawback of the proposed method, it is not possible to pick a parameter set that achieves very good adversarial and clean accuracy without losing generalizability to other problem sets. It is, however possible to tweak the hyperparameters to make either one of those stats look very good, adding a lot of noise, quantizing very far down and using a large amount of diffusion

iterations gives a good adversarial accuracy score against strong attacks but sacrifices clean accuracy. Upping quantization and adding less noise results in a good clean accuracy but strong attacks can reduce adversarial accuracy below 60%.

It is however possible to build the best set of hyperparameters on average, across experiments conducted, each hyperparameter choice is justified below.

The DDIM [38] sampler max timesteps $T$, are set using the following formulation $T = T_{max}//(t_{initial}//n)$, where $t_{initial}$ denotes the largest time step during denoising and $n$ denotes the number of diffusion steps. This convention was chosen to mirror Li et al [24], the reasoning being that $t_{initial}//n$ represents the amount of noise that is to be purged at every step, and then following the convention of optimized schedulers dividing their max timesteps by the actual amount of steps to be performed.

Tuning $t_{max}$ is the most important choice to make, as it controls the max amount of information the diffusion process can remove. Setting it too high kills clean accuracy but setting it too low allows adversarial information to propagate down the purification chain. Standard PGD attacks can be successfully defended with $t_{max}$ as low as 10, while C&W + BPDA + EOT requires up to 100, luckily setting this parameter higher than it has to be only slightly reduces clean accuracy as long as the number of diffusion steps is also increased to match. This makes sense, as a longer run of the generative process is needed to compensate for more data being destroyed. A value of 100 is chosen then to have the best chance on average at defending against both strong and weak attacks.

The number of timesteps $n$ was set to 100 due to compute constraints, a higher value might have increased adversarial accuracy but I was not able to test this on the hardware I had access to. Less than 100 diffusion steps results in strong attacks achieving $\leq$65% accuracy however, as shown in Figure 4.2.

The radius at which to perform frequency band blending (explained in section 3.1.1) was a very important hyperparameter to choose, this hyperparameter controls the ratio of low to high frequency data that is taken from $q(x_{adv})$ and $x_0$, where $q$ represents quantization, $x_{adv}$ and $x_0$ are the adversarial and clean prediction data points. If this radius is too low, $x_{guide}$ converges to $x_0$, and the algorithm behaves as a worse version version of DiffPure + $x_0$ voting, as the guidance system draws towards $x_0$, destroying clean accuracy. If this value is too high, $x_{guide}$ converges onto the quantized adversarial data point, almost directly feeding perturbations into the classifier. The value chosen was 0.1, with values of 0.1-0.2 working better for different attacks, 0.1 worked best on average. This value can be learned from data, which results in an optimal value for that specific attack type, however to ensure generalizability this value was manually tuned.

The quantization level chosen depends on the number of total steps, as a large number of steps means many unique $x_0$ predictions, as so a higher clean accuracy at the end. If the number of diffusion steps is 10-30, quantization can't go below 16/255 without killing clean accuracy, at higher diffusion steps, 60-100, quantizing even down to 6/255 retains clean accuracy $\geq$75%. Ultimately to ensure best average performance, a quantization level of 6 was chosen at at 100 diffusion steps.

The guidance scale parameter has to be chosen to keep the data point on the right diffusion trajectory without overwhelming it, the guidance gradient is $\ell_2$ normalized. A value of 1 was chosen first, then upped to 2 as heavier biasing improved robust accuracy of strong attacks at the cost of a bit of clean accuracy.

## 4.2 Standard Attacks

For each standard attack test, the full proposed method is tested against DiffPure [23], and DiffPure with $x_0$ estimate majority voting. Majority $x_0$ voting is enough to significantly improve upon DiffPure, and so by testing these three separately it is possible to see where the suggested modifications actually improve robustness. DiffPure is included to provide a baseline for diffusion based purification to improve upon, as is standard in literature.

Tested first, the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD).

FGSM involves simply adding a one step prediction in the sign of the direction which most increases loss, $x_{adversarial} = x + \epsilon \mathbf{sign}(\frac{\partial x}{\partial \text{loss(x)}})$. Iterating towards this direction and projecting back into the set

of allowed perturbations describes Projected Gradient Descent (PGD), a more effective but compute heavy attack algorithm. Finally, Ultimate PGD (UPGD) extends classical optimization technology improvements to the process, employing momentum, random restarts and specialised loss functions including C&W style margin loss [21].

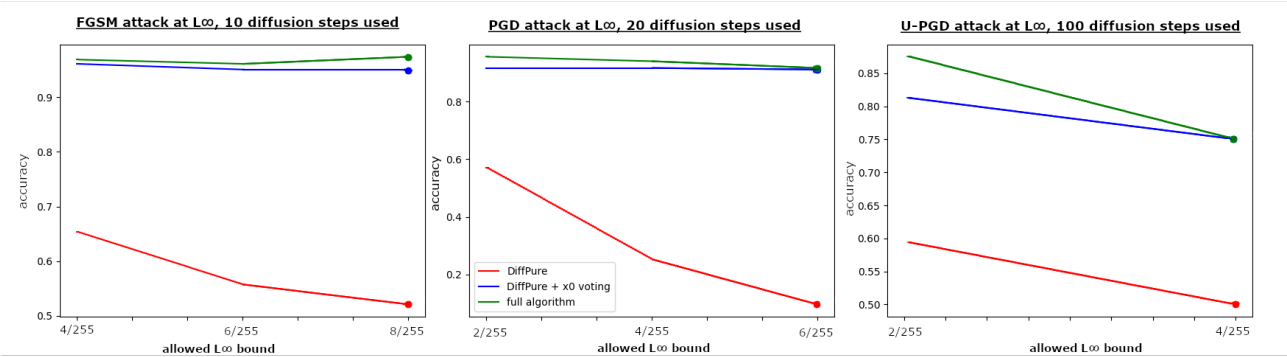


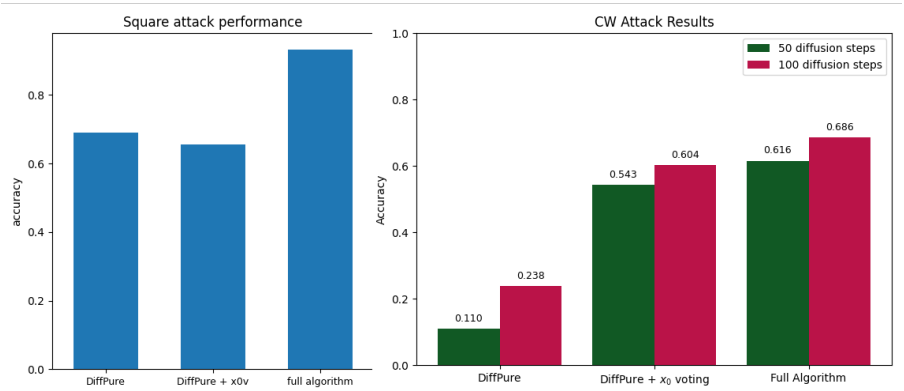

Figure 7: Adversarial performance of DiffPure [23] against DiffPure with $x_0$ estimate majority voting and the full proposed algorithm. Tested against n=512 data points using a ResNet-56 [59] classifier post purification. Note that for these tests, $t_{initial}$ as set a little lower than for further attacks, this was to show upper limits of accuracy against weak attacks, for best generalizability $t_{initial}$ should be set to 100, which would limit the clean and robust accuracies here to about 80%.

The full proposed algorithm, as well as DiffPure with $x_0$ voting drastically outperform regular DiffPure. Against the FGSM, the accuracy of DiffPure falls rapidly as $\epsilon$ approaches 8/255. This test is somewhat biased against regular DiffPure however, as DiffPure requires up to 100 diffusion steps to achieve peak purification, these tests used 10 and 20 steps for the first two experiments. Even so, the proposed method outperforms DiffPure at a fraction of the iteration count against these standard attacks. The full algorithm does provide a small but consistent advantage over just majority voting, but the attacks are not yet strong enough where this is significant. These attacks are all performed against an $\ell_\infty$ norm bound using straight forward, loss gradient methods. As explained in section 3.1, this encourages high frequency perturbations that $x_0$ estimates excel purging when combined with majority voting. These results support this claim, as there is little improvement from the rest of the proposed method, which focuses on perturbations that might survive in low frequency space. Going left to right, the attack strength increases, and so does the difference in performance between DiffPure + $x_0$ voting and the full algorithm, this follows from previous reasoning, $x_0$ voting is sufficient to purge weak, low level adversarial purifications. The FGSM generates only such perturbations and these two algorithms attain almost identical performance, whereas for U-PGD that gap widens as perturbations follow more complex patterns that aren't as easily purged just by averaging out mean projections towards the clean data manifold learned by diffusion.

Tested next, the Carlini Wagner (C&W) [21] and Square [40] attacks. The C&W attacks works by first utilizing a change of variables, setting $x + \epsilon = \frac{1}{2}(\tanh(W) + 1)$, where $\epsilon$ is the perturbation added to image $x$, and optimizing over $W$ directly. This form naturally clips the valid output over all values of $W$ to (0, 1), the bounds of an image, without harsh projection.

Carlini et al then optimize to minimize $||\epsilon||_2 + c \cdot f(x + \delta)$, where $f$ denotes a specialised objective function and $c$ is a constant balancing misclassification and perturbation magnitude. The function $f$ used is known as margin loss and is given by $f(x) = \max(Z(x)_{correct} - \max(Z(x)_{incorrect}), -\kappa)$, where $Z(x)_{correct}, Z(x)_{incorrect}$ denote the correct and incorrect logit given $x$, and $\kappa$ is a hyperparameter controlling how strong the misclassification aims to be. The loss function focuses only on minimizing the correct logit while maximizing the second most likely logit, while $\kappa$ is used to cut off optimization energy to logits that are already very negative. This optimization problem is then handed off to the ADAM [51] optimizer, a culmination of recent advancements in optimization technology. Generally, the C&W attack is strong as it reformulates the problem of finding a minimal adversarial examples

into a form best suited for modern optimizers, then focusing all optimization effort into just changing the relative positions of the most relevant logits.




A well tuned C&W + BPDA + EOT attack actually reduces the proposed method adversarial accuracy down to about 35%-55%, depending on defence hyperparameters, however the goal of the proposed method is to purify perturbations that are imperceptible to people. Perturbations with an $\ell_2$ norm above 0.75 are discarded, and even if the data point generated is adversarial it does not count towards adversarial accuracy. About a third of the data points generated were above this threshold. The confidence ($C$) parameter of the C&W attack was extensively tuned to perturb as many data points as possible without crossing over the disqualification boundary, usually, $C$ is found algorithmically by binary search, however in this case it resulted in a large number of data points being disqualified though they were technically adversarial. The full algorithm shows about an 8% improvement over DiffPure + $x_0$ voting.

The Square attack [40], instead operates under different assumptions, it is a black box attack so it does not use gradient information like the other attacks tested. Testing against black box attacks is crucial as a white box problem statement is a superset of the black box problem statement, just because an attacker has information about the inner working of an algorithm does not mean they have to use it. Furthermore, testing against black box attacks helps make sure that an adversarial defence is not only secure because of gradient masking or obfuscation. If black box attack success rate greatly exceeds white box success rate, it might also mean that the defence is over-fitting to the attack being tested.

The square attack works by attempting to place nearly transparent squares over various parts of the image, starting with larger squares to build a score of what parts of the image the classifier is most sensitive to. By then shrinking the sizes of the squares and concentrating efforts on the most sensitive parts of the image, the algorithm is able to efficiently generate harmful low frequency perturbations.

Interestingly, DiffPure + $x_0$ estimate voting performs worse than regular DiffPure, this is the only attack tested which has this result. This is likely because relying on $x_0$ estimates makes a defence excel at purging high frequency alterations while being very vulnerable to low frequency adversarial data that propagates through $x_0$ estimates. Regular DiffPure, however, does not pass these $x_0$ estimates to the classifier and so is only normally vulnerable, which in this case is less than DiffPure + $x_0$ voting.

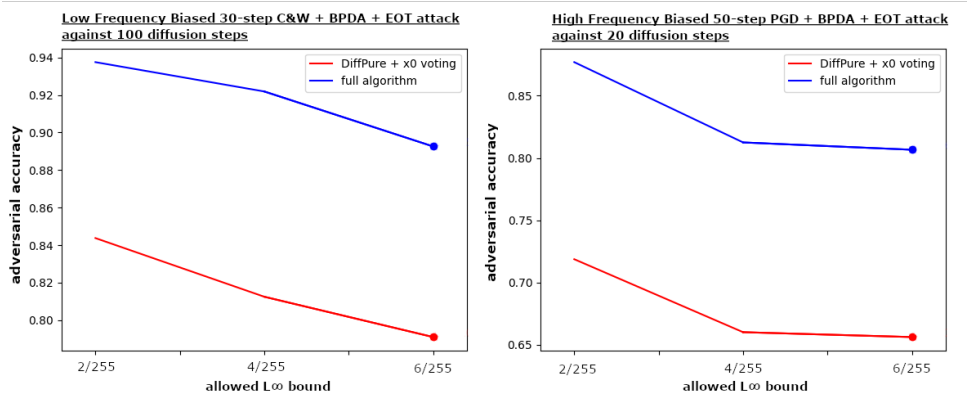
### 4.3 Adaptive Attacks

As justified in section 3.1, adversarial perturbations have a bias towards high frequency space, this motivates many aspects of the proposed design such as relying on high frequency data from $x_0$ estimates and injecting Gaussian noise when performing reverse diffusion. However the proposed method must be robust under attacks that make use of a systems over-reliance on this property for adversarial defence.

Inspired by the work of Guo et al who optimize for low frequency perturbations by perturbing low frequency cosine transform coefficients [47], I adapted a C&W [21] attack to optimize the DFT coefficients, real and imaginary parts separately. By projecting into frequency space, optimizing over low frequency components then projecting back into pixel space and clipping into allowed bounds, it is possible to generate effective low frequency perturbations. The C&W attack was chosen here as it

naturally produces lower frequency perturbations compared to PGD and other loss gradient methods, being an optimizer based method also made it easier to modify the space over which it learns, directly optimizing in DFT coefficient space instead of manually clipping away undesired parts of the gradient.

Naturally, it makes sense to test an attack of the opposite type, instead of forcing data into low frequencies, prioritize high frequencies. The obvious base attack to use is PGD, which already naturally produces high frequency data. I follow a methodology similar to Luo et al [64], who impose a low frequency deviation loss on the DWT decomposition of the perturbed image, forcing the optimization energy into high frequency components. Instead I explicitly zero out low frequency components of a 2D-DFT decomposition of the current perturbation at each PGD iteration, ensuring that when the perturbation is added, it can only contribute to low frequency components. Luo et al also use a C&W attack style base, I found that for this more simple, $\ell_\infty$ norm, bound problem, PGD reduces robustness further, likely due to the already high frequency nature of PGD perturbations. C&W perturbations are in general more spread out among frequency bands, so my explicit zeroing out hurts more, this is a limitation of my implementation however, not one inherent to the C&W attack, as others have seen success biasing this attack towards high frequency components [65], though this was done against an $\ell_2$ norm bound.




The gap between the results of DiffPure + $x_0$ voting and the full algorithm is now up to 15% at an $\ell_\infty, 3/255$ norm bound, proving that improvements past $x_0$ majority voting provide genuine adversarial robustness against high frequency attacks.

More interestingly, this algorithm, combined with optimal BPDA placement and 4 EOT iterations, reduces the effectiveness of DiffPure + $x_0$ voting all the way down to 65%. This large dip in performance compared to previous testing implies that $x_0$ voting is not good at purging more complex, adapted high frequency perturbations, not just low frequency ones. However, as shown above in section 4.2, it is enough against simple ones.

### 4.4 Direct Comparison to Similar Works

As discussed, many elements of the proposed method were inspired and adapted from similar literature, the frequency band blending was adapted from Pei et al and their work on FreqPure [35], with the implementation I settled upon being slightly more simple, they perform more complex separate amplitude and phase spectra filtering. The guidance formulation was very similar to the one used by Wang et al and their work on GDMP, just an analytic simplification under simple assumptions. Both these choices are further justified throughout section 3.

To show that the proposed method offers tangible improvements over these works, this section presents a direct empirical comparison against these two methods. I implemented both in Pytorch, closely following the methodology of the original papers

First the methods are tested against a classic C&W attack + optimal BPDA + EOT (4) against a $\ell_2$ norm bound, as well as against PGD + optimal BPDA + EOT (4). They all use only 50 diffusion steps in order to more clearly show relative differences, at higher step counts the algorithms all perform more similarly.

The attacks perform far more similarly against the PGD attack, even with EOT and BPDA, PGD

is fundamentally limited by its optimizer, it attempts to optimize for a large loss using only Stochastic Gradient Descent (SGD) which is almost always strictly weaker than ADAM [51] in this context.

The proposed method outperformed GDMP and FreqPure in the tests, however against PGD, the results of the full algorithm and FreqPure were within worst case error bars of each other. There was not enough evidence to conclude that the proposed method outperforms FreqPure against $\ell_2$ PGD. Furthermore FreqPure achieved a clean accuracy of 96% in these tests, GDMP achieved 91%, while the full algorithm achieved only 83%. The proposed method does offer slightly better adversarial performance at a significant drop in clean accuracy.

Repeating this experiment against larger allowed norm bounds also results in the other algorithm performing better, specifically at $\ell\infty$ bounds near 8/255, the performance gap between FreqPure and the proposed method closes, with FreqPure performing better at bounds above 10/255. The proposed method offers an edge only when purifying imperceptible perturbations.

### 4.5 Miscellaneous Testing

In order to claim generalisability of the proposed method, it has to be shown working on different data sets. The proposed method was tested on the MNIST, a more simple data set. Unsurprisingly, the proposed method performs very well, with a C&W attack against MNIST failing to reduce robust accuracy below 90%.

| Attack | DiffPure | DiffPure + x0v | Full algorithm |
|---|---|---|---|
| 50 step CW + BPDA L2 | 0.969 | 0.969 | 0.965 |
| 50 step PGD + BPDA L2 | 0.984 | 0.992 | 0.965 |




Figure 8: MNIST adversarial performance against an $(\ell_2, 0.75)$ norm bound, n=256 data points tested for 30 diffusion steps, very large transformer classifier used. Also pictured, complexity comparison between MNIST and fashion-MNIST.

Something very interesting happens, however, when attempting to use a MNIST diffusion model to purify perturbations on the fashion-MNIST dataset, a dataset of equal resolution but greater complexity. Attempting this using CIFAR-10 hyperparameters results in robust performance under 15%, barely better than a random guesser, however lowering $t_{initial}$ to 10, using only 50 diffusion steps, dropping guidance by a factor of 10 and using a transformer classifier backend, first trained on Imagenet then fine tuned on MNIST leads to a very interesting result.

Even though the diffusion model used here has never seen a single fashion-MNIST data point, DiffPure + $x_0$ voting achieve 50.7% adversarial accuracy against 30 step U-PGD $(\ell_\infty, 16/255)$! Note that the same attack reduced DiffPure + $x_0$ voting down to nearly 65% (see Figure 8), when using a diffusion model trained on the exact distribution it is being tested against.

**Zero-Shot Diffusion Purification Performance with Different Classifiers**

| | DiffPure + x0v + FCN | DiffPure + x0v + CNN | DiffPure + x0v + TRS | Full algorithm | DiffPure | TRS |
|---|---|---|---|---|---|---|
| 30 step U-PGD + BPDA L∞ 16/255 | 0.395 | 0.449 | **0.508** | 0.354 | 0.292 | 0 |

This security is a result of the combination of the DiffPure + $x_0$ voting algorithm and the transformer classifier (denoted TRS in the table above), as alone U-PGD reduces both their adversarial accuracies to 39% and 0% respectively. Repeating this experiment with fully connected and convolutional classifiers results in significantly worse performance, the full algorithm and DiffPure also fail to adapt to this circumstance, though still perform better than a random guesser. This experiment indicates that diffusion models to some degree generalize beyong their training data as adversarial purifiers perhaps learning a more general clean data manifold. Practically this is strong evidence for a

diffusion based purifier completely decoupled from the classifier data set being possible, further testing needs to be done on this area.

To analyse practical applicability, the compute time per data point was measured against batch size and diffusion step count. The test was performed using CUDA parallelization, and in software on a CPU, at least 2gb of video and system memory was kept clear during these tests to minimize paging and the device was left idle when testing to avoid taking up meaningful CPU time.

the results are shown in appendix section .1.1. It is clear from this graph that compute time is dominated by noise prediction evaluations, as the relationship between compute time and diffusion steps is linear even at small timesteps. This analysis also shows that increasing batch size increases performance per data point, this makes sense as larger tensors allow the use of algorithms with better time complexities, the graph also makes it clear that increasing batch size does not hit diminishing returns even for 256 data points.

# 5 Further Discussion and Conclusions

## 5.1 Summary of Findings

Through rigorous testing and evaluation, the strengths and weaknesses as well as practical limitations of the proposed method have been studied. The project succeeds in producing a method for purging very small, imperceptible alterations to an image better than similar work in limited situations. The method adapts well to unseen circumstances using the same hyperparameter set, however does not achieve state of the art performance without hyperparameter tuning, which would defeat the point of a transferable defence. The method has been shown to be secure against perturbations focused into both high and low frequency bands, and against strong adaptive attacks that make use of BPDA and EOT compensation.

The proposed method however suffers from a poor clean accuracy and worse purification results than similar works, a limitation of prioritizing adaptability over strict performance in any one bound. The method is also very computationally expensive.

A far simpler algorithm, one performed by simply majority voting over $x_0$ estimates computed at every denoising step achieves similar performance to the full proposed method. However, this process is very vulnerable to low frequency perturbations, even more vulnerable than DiffPure [23], an even simpler algorithm.

## 5.2 Critical Analysis of Aim Completion

### 5.2.1 Aim 1

Aim 1 (Design a diffusion-based adversarial purification framework) was met; the proposed method makes use of all three of the components described. Frequency band blending (1) is used to isolate high and low frequency data for it to be purified separately, the result is the target for classifier free guidance (2) and actual purification happens via diffusion. The design was implemented in Pytorch and verified to work against both CIFAR-10 and MNIST data sets, I have built the full automatic pipeline that, given a classifier and diffusion model, is able to perform reliable adversarial purification and classification.

Different surrogates were tested for both the low frequency perturbation resistant function and for the frequency band blending. For the first, local contrast normalization, randomized smoothing, quantization and a smooth approximation of contrast limited adaptive histogram equalization filtering was tested. Quantization achieved best results on average, leading to it being chosen for this purpose. Discrete wavelet transforms, regular discrete Fourier transforms as well as row-PCA [**?**] were tested as potential back-ends for combining the frequency data of two images. Row-PCA performed by far the worst, being reduced to about 30% adversarial accuracy, it also took about twice as long to compute

each reconstruction. DWT and 2D-DFT performed very similarly, however the 2D-DFT seemed to perform strictly better, the reasons for this are discussed in section 3.1.1.

#### 5.2.2 Aim 2

Aim 2 (achieve strong robustness against standard white-box attacks) was met. The proposed method achieved ≥90% accuracy against the FGSM and ≥85% accuracy against PGD, both above the 80% accuracy goal. The proposed method achieved compensated 68% accuracy against C&W + BPDA + EOT and ≥93% performance against Square. While C&W did reduce the strict robust accuracy further, it was not able to do so in a way that kept them imperceptibly different, the confidence $C$ and $\kappa$ parameters were extensively tuned to ensure best performance here.

#### 5.2.3 Aim 3

Aim 3 (Evaluate robustness under advanced adaptive attacks) was met. Different BPDA placement was tested to ensure reported robustness is due to actual security not obfuscation, a non-standard BPDA placement was found to reduce robust accuracy furthest, one over only the quantization. EOT iterations were progressively increased until they stopped providing benefit, which was at 4 iterations, this value was used for further testing to prove security even under averaged out gradients. The algorithm was tested against C&W and PGD variants with BPDA and EOT, robust accuract remained above the 60% target in all cases.

A custom attack was designed to generate only high frequency perturbations, this attack was tested against the proposed method, which still managed to achieve 79% robust accuracy.

#### 5.2.4 Aim 4

Aim 4 (Directly compare against similar algorithms) was met. The algorithm was compared against four similar algorithms, being compared against DiffPure and DiffPure + $x_0$ voting for many experiments. These two attacks became the baseline for judging algorithmic improvement. The method was next directly compared against two published works FreqPure [35] and GDMP [29] and was found to achieve better performance in a very specific problem bound. However, FreqPure still beat the proposed method in clean accuaracy and robust accuracy at higher perturbation magnitudes.

#### 5.2.5 Stretch Goals

The proposed method was tested against FreqPure and GDMP, and the places where the proposed method is better and worse were found, the attack was tested and acheived good performance against low frequency perturbations. The method was tested against othe data sets but not more complex ones. a CIFAR-10 diffusion model was shown to posess some zero shot purification ability.

### 5.3 Overall Strength and Weaknesses of Proposed Method

The proposed algorithm excels against attacks that rely on concentrating perturbations into narrow frequency bands, as shown it achieved far higher robust accuracy than DiffPure + $x_0$ voting when compared against adaptive algorithms that attempt to build perturbations in only high or low bands. The proposed method describes a robust classification algorithm that adapts well to unforeseen norm bounds, the proposed method is secure against both white and black box attacks as long as the perturbations are not of large magnitude.

The difference in performance between DiffPure, DiffPure + $x_0$ voting and the full algorithm is largest when tested against these adaptive attacks. This shows that the proposed method genuinely does provide non-trivial robustness improvements against non-standard attacks. The proposed method was shown to be secure even under optimal BPDA, and sufficient EOT iterations implying that

robustness is due to security not obfuscation. The good score against the black box Square attack supports this, as robust accuracy did not significantly drop compared to other attacks.

However, the proposed method trades adaptability for performance to some degree, instead of attempting to get the best possible score on one or two tests, the method was designed to work best on average over many attack types and bounds. A consequence of this is worse individual performance, many purification schemes exist that can beat a 68% C&W attack score without breaking an $\ell_2$ norm of 0.75. Furthermore, in order to achieve a good robust accuracy, this algorithm sacrifices clean accuracy, for which it gets a significantly worse score than FreqPure and GDMP. Finally, the algorithm is very computationally expensive, in terms of both compute and memory. Just the parameters required for the diffusion model take up over half a gigabyte, with memory usage routinely exceeding 1 gigabyte for extended purification runs.

### 5.4 Future Work

Quantization performing better than the other low frequency perturbation resistant functions was an unexpected and somewhat disappointing result. Testing more candidate functions is required, perhaps ones based on spectral alignment regularization [66], aligning low frequencies against a set of clean examples for example might result in a function that works a lot better.

Another possible idea for this function would be one based on Discrete Cosine Transform (DCT) filtering, potentially using the 2D-DFT to pass of data points to the majority voting scheme while creating targets for guidance with DCT filtering. An attacker would then have to perturb a data point such as to fool both the DCT and 2D-DFT filters.

The method should be tested against more data sets, MNIST and CIFAR-10 are trivial data sets in the modern era and proper evaluation would involve testing against Imagenet or similar high resolution data sets. Testing on MNIST after designing a method to work for CIFAR-10 resulted in the strongest attacks tested being unable to break 90% robust accuracy. It follows that this method would probably not work well against higher resolution, more complex data sets, this direction would be good to explore in future work.

Finally, the limits of zero show purification should be explored, maybe to develop a single diffusion model apt at any image data type.

### 5.5 Conclusion

This study presents and evaluates a novel diffusion-based purification framework which exhibits very good performance against strong adaptive attacks that attempt to force adversarial energy into just a few frequency bands. It excels at purifying low magnitude, imperceptible perturbations however falls behind similar methods when perturbation magnitude increases. In the study of the proposed method a far simpler method was revealed, simply performing regular DiffPure style purification, at each step predicting $x_0$ and passing that value off to the classifier for a majority vote. This method achieves very close to the performance of the full method at a fraction of the complexity in all but very specific, adaptive attacks.

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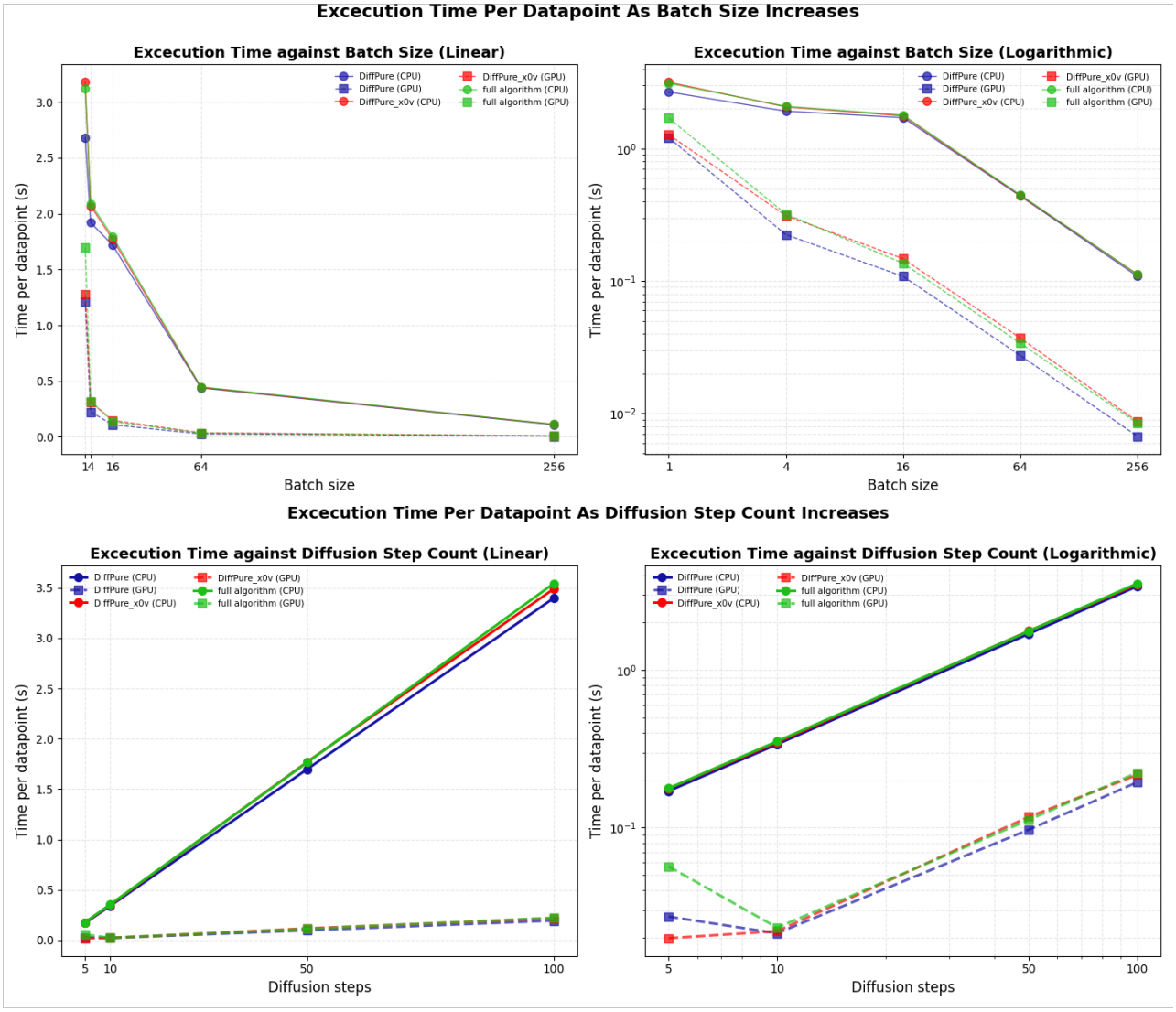
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# Appendix A

## .1 Additional Graphs

### .1.1 Execution Time Against Batch Size and Diffusion Step Count



## .2 Similar Attack Comparison

[!h]

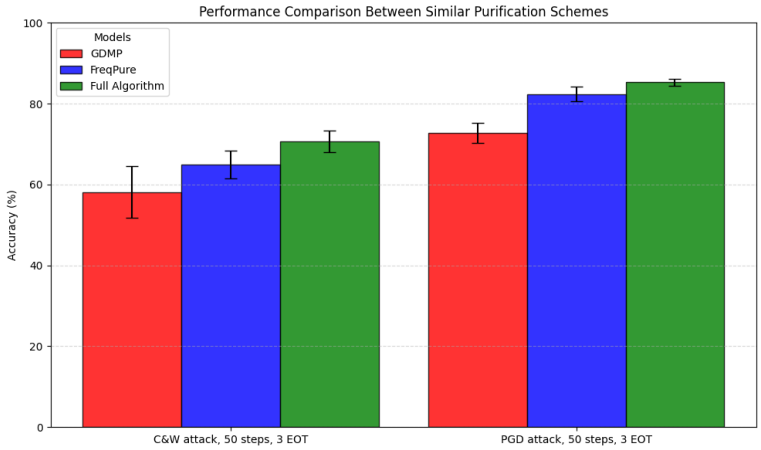



Figure 9: Adversarial accuracy results compared to similar methods, test repeated 8 times with 128 data points each time, 1024 CIFAR-10 data points total. Range between worst and best case result plotted as error bars.